

Mausam Rana

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Data Analyst with a background in software engineering and advanced analytics, experienced in transforming complex datasets into actionable insights using SQL, Python, R, and Power BI. Skilled in predictive modeling, ETL pipelines, and interactive dashboard development to support data-driven decision making. Proven ability to automate reporting workflows, build machine learning models, and deliver scalable analytics solutions that improve operational efficiency and business insights.

TECHNICAL SKILLS

Core Analytics: SQL, Python, R, Statistical Analysis (Regression, Hypothesis Testing, A/B Testing)

Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, Data Preprocessing, Feature Engineering, Model Evaluation

BI & Data Visualization: Power BI (DAX), Tableau, Streamlit, R Shiny, Excel (Advanced Functions, Pivot Tables, VBA)

Cloud & Databases: Snowflake, AWS, Microsoft SQL Server, MySQL, PostgreSQL, MongoDB

Automation & Collaboration Tools: Power Automate, Salesforce Flow, Git, Confluence, JIRA

Development & APIs: MERN Stack (MongoDB, Express.js, React.js, Node.js), REST APIs, JavaScript

Analytics Methods: ETL Pipelines, Data Modeling, Predictive Analytics, A/B Testing, KPI Development, Data Governance

PROFESSIONAL EXPERIENCE

Data Analyst, Impact Initiative Nepal

April 2024 - July 2024

- Developed predictive analytics models using **Python and R** to forecast student retention and salary trends, enabling improved academic planning and data-driven decision making.
- Built **interactive Power BI dashboards** visualizing enrollment patterns, financial aid distribution, student performance metrics, improving institutional insights, contributing to a **10% improvement in retention analysis accuracy**.
- Automated reporting workflows using **SQL queries and Python scripts**, reducing manual reporting time by **25%** and improving reporting reliability across multiple datasets.
- Performed **exploratory data analysis (EDA), statistical testing, and text classification** to analyze student feedback and institutional datasets for strategic planning initiatives.

Software Developer, Beehives Technology

July 2019 - January 2021

- Built **data-driven web applications using the MERN stack (MongoDB, Express, React, Node.js)**, integrating REST APIs and database queries to support real-time analytics and reporting features.
- Designed and optimized **MongoDB database schemas and aggregation queries**, improving large dataset retrieval performance for application analytics.
- Developed backend **data processing pipelines using Node.js**, transforming and storing structured application data from multiple API sources.
- Implemented **data validation and transformation logic** ensuring high data integrity across user activity and transactional datasets.
- Optimized database queries and application data handling, improving system performance and scalability for large user datasets.

PROJECTS

Location-Based Sales Forecasting & Interactive Dashboard

October 2025

- Developed a sales prediction model using Snowpark ML pipelines on Snowflake with ETL workflows and built interactive Power BI and Streamlit dashboards for real-time visualization of historical trends and location-based recommendations.

Data Analytics Project - BI Solution | Seed Stage Startup

April 2025

- Performed EDA and designed ML classification models (SVM, Random Forest) achieving 83% accuracy on 250+ records, creating Power BI and R Shiny dashboards for KPI tracking improving lead generation.
- Built predictive models to detect diabetes using diagnostic health data**

CERTIFICATION

- Microsoft Certified: Power BI Data Analyst Associate (PL-300)** - In Progress
- CS50's Web Programming with Python and JavaScript**, Harvard University
- "Data Science with Python"**- Broadway Infosys Institute

EDUCATION

University of New Haven, Master of Science in Analytics (Deans Scholarship)

Connecticut, USA

Coursework: Data Mining, Machine Learning, Data Visualization, Database Management, Business Forecasting

GPA 4.0

Westcliff University, MBA in Technology and Innovation (Full Scholarship)

Irvine, California

GPA 4/4

Nitte Meenakshi Institute of Technology, India, Bachelor of Engineering in Computer Science

Bangalore, India